78XX

# 3-Terminal 1A Positive Voltage Regulator

## Features

- Output Current up to 1A
- Output Voltages of 5, 6, 8, 9, 10, 12, 15, 18, 24
- Thermal Overload Protection
- Short Circuit Protection
- Output Transistor Safe Operating Area Protection

## General Description

The LM78XX series of three terminal positive regulators are available in the TO-220 package and with several Þxed output voltages, making them useful in a wide range of applications. Each type employs internal current limiting, thermal shut down and safe operating area protection, making it essentially indestructible. If adequate heat sinking is provided, they can deliver over 1A output current. Although designed primarily as Þxed voltage regulators, these devices can be used with external components to obtain adjustable voltages and currents.

## Ordering Information

| Product Number | Output Voltage Tolerance | Package | Operating Temperature |
|---|---|---|---|
| LM7805CT | ±4% | TO-220 | -40¡C to +125¡C |
| LM7806CT | | | |
| LM7808CT | | | |
| LM7809CT | | | |
| LM7810CT | | | |
| LM7812CT | | | |
| LM7815CT | | | |
| LM7818CT | | | |
| LM7824CT | | | |
| LM7805ACT | ±2% | | 0¡C to +125¡C |
| LM7806ACT | | | |
| LM7808ACT | | | |
| LM7809ACT | | | |
| LM7810ACT | | | |
| LM7812ACT | | | |
| LM7815ACT | | | |
| LM7818ACT | | | |
| LM7824ACT | | | |

## Block Diagram

Figure 1.

## Pin Assignment

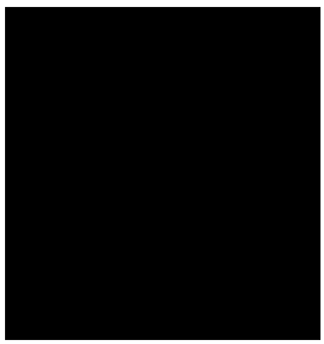


Figure 2.

## Absolute Maximum Ratings

Absolute maximum ratings are those values beyond which damage to the device may occur. The datasheet speciÞcations should be met, without exception, to ensure that the system design is reliable over its power supply, temperature, and output/input loading variables. Fairchild does not recommend operation outside datasheet speciÞcations.

| Symbol | Parameter | | Value | Unit |
|---|---|---|---|---|
| $V_I$ | Input Voltage | $V_O$ = 5V to 18V | 35 | V |
| | | $V_O$ = 24V | 40 | V |
| $R_{JC}$ | Thermal Resistance Junction-Cases (TO-220) | | 5 | ¡C/W |
| $R_{JA}$ | Thermal Resistance Junction-Air (TO-220) | | 65 | ¡C/W |
| $T_{OPR}$ | Operating Temperature Range | LM78xx | -40 to +125 | ¡C |
| | | LM78xxA | 0 to +125 | |
| $T_{STG}$ | Storage Temperature Range | | -65 to +150 | ¡C |

## Electrical Characteristics (LM7805)

Refer to the test circuits. -40¡C < $T_J$ < 125¡C, $I_O$ = 500mA, $V_I$ = 10V, $C_I$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 4.8 | 5.0 | 5.2 | V |
| | | 5mA $I_O$ 1A, $P_O$ 15W, $V_I$ = 7V to 20V | | 4.75 | 5.0 | 5.25 | |
| Regline | Line Regulation[1] | $T_J$ = +25¡C | $V_O$ = 7V to 25V | Ð | 4.0 | 100 | mV |
| | | | $V_I$ = 8V to 12V | Ð | 1.6 | 50.0 | |
| Regload | Load Regulation[1] | $T_J$ = +25¡C | $I_O$ = 5mA to 1.5A | Ð | 9.0 | 100 | mV |
| | | | $I_O$ = 250mA to 750mA | Ð | 4.0 | 50.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.0 | 8.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | 0.03 | 0.5 | mA |
| | | $V_I$ = 7V to 25V | | Ð | 0.3 | 1.3 | |
| $V_O$/ T | Output Voltage Drift[2] | $I_O$ = 5mA | | Ð | -0.8 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 42.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection[2] | f = 120Hz, $V_O$ = 8V to 18V | | 62.0 | 73.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance[2] | f = 1kHz | | Ð | 15.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 230 | Ð | mA |
| $I_{PK}$ | Peak Current[2] | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

1. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
2. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7806) (Continued)

Refer to the test circuits. -40¡C < $T_J$ < 125¡C, $I_O$ = 500mA, $V_I$ = 11V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 5.75 | 6.0 | 6.25 | V |
| | | 5mA $I_O$ 1A, $P_O$ 15W, $V_I$ = 8.0V to 21V | | 5.7 | 6.0 | 6.3 | |
| Regline | Line Regulation[3] | $T_J$ = +25¡C | $V_I$ = 8V to 25V | Ð | 5.0 | 120 | mV |
| | | | $V_I$ = 9V to 13V | Ð | 1.5 | 60.0 | |
| Regload | Load Regulation[3] | $T_J$ = +25¡C | $I_O$ = 5mA to 1.5A | Ð | 9.0 | 120 | mV |
| | | | $I_O$ = 250mA to 750mA | Ð | 3.0 | 60.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.0 | 8.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 8V to 25V | | Ð | Ð | 1.3 | |
| $V_O$/ T | Output Voltage Drift[4] | $I_O$ = 5mA | | Ð | -0.8 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 45.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection[4] | f = 120Hz, $V_O$ = 8V to 18V | | 62.0 | 73.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance[4] | f = 1kHz | | Ð | 19.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current[4] | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

3. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
4. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7808) (Continued)

Refer to the test circuits. -40¡C < $T_J$ < 125¡C, $I_O$ = 500mA, $V_I$ = 14V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 7.7 | 8.0 | 8.3 | V |
| | | 5mA $I_O$ 1A, $P_O$ 15W, $V_I$ = 10.5V to 23V | | 7.6 | 8.0 | 8.4 | |
| Regline | Line Regulation(5) | $T_J$ = +25¡C | $V_I$ = 10.5V to 25V | Ð | 5.0 | 160 | mV |
| | | | $V_I$ = 11.5V to 17V | Ð | 2.0 | 80.0 | |
| Regload | Load Regulation(5) | $T_J$ = +25¡C | $I_O$ = 5mA to 1.5A | Ð | 10.0 | 160 | mV |
| | | | $I_O$ = 250mA to 750mA | Ð | 5.0 | 80.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.0 | 8.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | 0.05 | 0.5 | mA |
| | | $V_I$ = 10.5V to 25V | | Ð | 0.5 | 1.0 | |
| $V_O$/ T | Output Voltage Drift(6) | $I_O$ = 5mA | | Ð | -0.8 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 52.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection(6) | f = 120Hz, $V_O$ = 11.5V to 21.5V | | 56.0 | 73.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance(6) | f = 1kHz | | Ð | 17.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 230 | Ð | mA |
| $I_{PK}$ | Peak Current(6) | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

5. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
6. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7809) (Continued)

Refer to the test circuits. -40¡C < $T_J$ < 125¡C, $I_O$ = 500mA, $V_I$ = 15V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 8.65 | 9.0 | 9.35 | V |
| | | 5mA $I_O$ 1A, $P_O$ 15W, $V_I$ = 11.5V to 24V | | 8.6 | 9.0 | 9.4 | |
| Regline | Line Regulation(7) | $T_J$ = +25¡C | $V_I$ = 11.5V to 25V | Ð | 6.0 | 180 | mV |
| | | | $V_I$ = 12V to 17V | Ð | 2.0 | 90.0 | |
| Regload | Load Regulation(7) | $T_J$ = +25¡C | $I_O$ = 5mA to 1.5A | Ð | 12.0 | 180 | mV |
| | | | $I_O$ = 250mA to 750mA | Ð | 4.0 | 90.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.0 | 8.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 11.5V to 26V | | Ð | Ð | 1.3 | |
| $V_O$/ T | Output Voltage Drift(8) | $I_O$ = 5mA | | Ð | -1.0 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 58.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection(8) | f = 120Hz, $V_O$ = 13V to 23V | | 56.0 | 71.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance(8) | f = 1kHz | | Ð | 17.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current(8) | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

7. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
8. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7810) (Continued)

Refer to the test circuits. -40¡C < $T_J$ < 125¡C, $I_O$ = 500mA, $V_I$ = 16V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 9.6 | 10.0 | 10.4 | V |
| | | 5mA $I_O$ 1A, $P_O$ 15W, $V_I$ = 12.5V to 25V | | 9.5 | 10.0 | 10.5 | |
| Regline | Line Regulation(9) | $T_J$ = +25¡C | $V_I$ = 12.5V to 25V | Ð | 10.0 | 200 | mV |
| | | | $V_I$ = 13V to 25V | Ð | 3.0 | 100 | |
| Regload | Load Regulation(9) | $T_J$ = +25¡C | $I_O$ = 5mA to 1.5A | Ð | 12.0 | 200 | mV |
| | | | $I_O$ = 250mA to 750mA | Ð | 4.0 | 400 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.1 | 8.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 12.5V to 29V | | Ð | Ð | 1.0 | |
| $V_O$/ T | Output Voltage Drift(10) | $I_O$ = 5mA | | Ð | -1.0 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 58.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection(10) | f = 120Hz, $V_O$ = 13V to 23V | | 56.0 | 71.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance(10) | f = 1kHz | | Ð | 17.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current(10) | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

9. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
10. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7812) (Continued)

Refer to the test circuits. -40¡C < $T_J$ < 125¡C, $I_O$ = 500mA, $V_I$ = 19V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 11.5 | 12.0 | 12.5 | V |
| | | 5mA $I_O$ 1A, $P_O$ 15W, $V_I$ = 14.5V to 27V | | 11.4 | 12.0 | 12.6 | |
| Regline | Line Regulation(11) | $T_J$ = +25¡C | $V_I$ = 14.5V to 30V | Ð | 10.0 | 240 | mV |
| | | | $V_I$ = 16V to 22V | Ð | 3.0 | 120 | |
| Regload | Load Regulation(11) | $T_J$ = +25¡C | $I_O$ = 5mA to 1.5A | Ð | 11.0 | 240 | mV |
| | | | $I_O$ = 250mA to 750mA | Ð | 5.0 | 120 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.1 | 8.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | 0.1 | 0.5 | mA |
| | | $V_I$ = 14.5V to 30V | | Ð | 0.5 | 1.0 | |
| $V_O$/ T | Output Voltage Drift(12) | $I_O$ = 5mA | | Ð | -1.0 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 76.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection(12) | f = 120Hz, $V_I$ = 15V to 25V | | 55.0 | 71.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance(12) | f = 1kHz | | Ð | 18.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 230 | Ð | mA |
| $I_{PK}$ | Peak Current(12) | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

11. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
12. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7815) (Continued)

Refer to the test circuits. -40¡C < $T_J$ < 125¡C, $I_O$ = 500mA, $V_I$ = 23V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 14.4 | 15.0 | 15.6 | V |
| | | 5mA $I_O$ 1A, $P_O$ 15W, $V_I$ = 17.5V to 30V | | 14.25 | 15.0 | 15.75 | |
| Regline | Line Regulation[13] | $T_J$ = +25¡C | $V_I$ = 17.5V to 30V | Ð | 11.0 | 300 | mV |
| | | | $V_I$ = 20V to 26V | Ð | 3.0 | 150 | |
| Regload | Load Regulation[13] | $T_J$ = +25¡C | $I_O$ = 5mA to 1.5A | Ð | 12.0 | 300 | mV |
| | | | $I_O$ = 250mA to 750mA | Ð | 4.0 | 150 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.2 | 8.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 17.5V to 30V | | Ð | Ð | 1.0 | |
| $V_O$/ T | Output Voltage Drift[14] | $I_O$ = 5mA | | Ð | -1.0 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 90.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection[14] | f = 120Hz, $V_I$ = 18.5V to 28.5V | | 54.0 | 70.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance[14] | f = 1kHz | | Ð | 19.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current[14] | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

13. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
14. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7818) (Continued)

Refer to the test circuits. -40¡C < $T_J$ < 125¡C, $I_O$ = 500mA, $V_I$ = 27V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 17.3 | 18.0 | 18.7 | V |
| | | 5mA $I_O$ 1A, $P_O$ 15W, $V_I$ = 21V to 33V | | 17.1 | 18.0 | 18.9 | |
| Regline | Line Regulation(15) | $T_J$ = +25¡C | $V_I$ = 21V to 33V | Ð | 15.0 | 360 | mV |
| | | | $V_I$ = 24V to 30V | Ð | 5.0 | 180 | |
| Regload | Load Regulation(15) | $T_J$ = +25¡C | $I_O$ = 5mA to 1.5A | Ð | 15.0 | 360 | mV |
| | | | $I_O$ = 250mA to 750mA | Ð | 5.0 | 180 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.2 | 8.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 21V to 33V | | Ð | Ð | 1.0 | |
| $V_O$/ T | Output Voltage Drift(16) | $I_O$ = 5mA | | Ð | -1.0 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 110 | Ð | µV/$V_O$ |
| RR | Ripple Rejection(16) | f = 120Hz, $V_I$ = 22V to 32V | | 53.0 | 69.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance(16) | f = 1kHz | | Ð | 22.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current(16) | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

15. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.

16. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7824) (Continued)

Refer to the test circuits. -40¡C < $T_J$ < 125¡C, $I_O$ = 500mA, $V_I$ = 33V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 23.0 | 24.0 | 25.0 | V |
| | | 5mA $I_O$ 1A, $P_O$ 15W, $V_I$ = 27V to 38V | | 22.8 | 24.0 | 25.25 | |
| Regline | Line Regulation[17] | $T_J$ = +25¡C | $V_I$ = 27V to 38V | Ð | 17.0 | 480 | mV |
| | | | $V_I$ = 30V to 36V | Ð | 6.0 | 240 | |
| Regload | Load Regulation[17] | $T_J$ = +25¡C | $I_O$ = 5mA to 1.5A | Ð | 15.0 | 480 | mV |
| | | | $I_O$ = 250mA to 750mA | Ð | 5.0 | 240 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.2 | 8.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | 0.1 | 0.5 | mA |
| | | $V_I$ = 27V to 38V | | Ð | 0.5 | 1.0 | |
| $V_O$/ T | Output Voltage Drift[18] | $I_O$ = 5mA | | Ð | -1.5 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 60.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection[18] | f = 120Hz, $V_I$ = 28V to 38V | | 50.0 | 67.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| rO | Output Resistance[18] | f = 1kHz | | Ð | 28.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 230 | Ð | mA |
| $I_{PK}$ | Peak Current[18] | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

17. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
18. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7805A) (Continued)

Refer to the test circuits. 0¡C < $T_J$ < 125¡C, $I_O$ = 1A, $V_I$ = 10V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 4.9 | 5.0 | 5.1 | V |
| | | $I_O$ = 5mA to 1A, $P_O$ 15W, $V_I$ = 7.5V to 20V | | 4.8 | 5.0 | 5.2 | |
| Regline | Line Regulation(19) | $V_I$ = 7.5V to 25V, $I_O$ = 500mA | | Ð | 5.0 | 50.0 | mV |
| | | $V_I$ = 8V to 12V | | Ð | 3.0 | 50.0 | |
| | | $T_J$ = +25¡C | $V_I$ = 7.3V to 20V | Ð | 5.0 | 50.0 | |
| | | | $V_I$ = 8V to 12V | Ð | 1.5 | 25.0 | |
| Regload | Load Regulation(19) | $T_J$ = +25¡C, $I_O$ = 5mA to 1.5A | | Ð | 9.0 | 100 | mV |
| | | $I_O$ = 5mA to 1A | | Ð | 9.0 | 100 | |
| | | $I_O$ = 250mA to 750mA | | Ð | 4.0 | 50.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.0 | 6.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 8V to 25V, $I_O$ = 500mA | | Ð | Ð | 0.8 | |
| | | $V_I$ = 7.5V to 20V, $T_J$ = +25¡C | | Ð | Ð | 0.8 | |
| $V_O$/ T | Output Voltage Drift(20) | $I_O$ = 5mA | | Ð | -0.8 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 10.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection(20) | f = 120Hz, $I_O$ = 500mA, $V_I$ = 8V to 18V | | Ð | 68.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance(20) | f = 1kHz | | Ð | 17.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current(20) | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

19. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
20. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7806A) (Continued)

Refer to the test circuits. 0¡C < $T_J$ < 125¡C, $I_O$ = 1A, $V_I$ = 11V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | 5.58 | 6.0 | 6.12 | V |
| | | $I_O$ = 5mA to 1A, $P_O$ 15W, $V_I$ = 8.6V to 21V | 5.76 | 6.0 | 6.24 | |
| Regline | Line Regulation(21) | $V_I$ = 8.6V to 25V, $I_O$ = 500mA | Ð | 5.0 | 60.0 | mV |
| | | $V_I$ = 9V to 13V | Ð | 3.0 | 60.0 | |
| | | $T_J$ = +25¡C, $V_I$ = 8.3V to 21V | Ð | 5.0 | 60.0 | |
| | | $T_J$ = +25¡C, $V_I$ = 9V to 13V | Ð | 1.5 | 30.0 | |
| Regload | Load Regulation(21) | $T_J$ = +25¡C, $I_O$ = 5mA to 1.5A | Ð | 9.0 | 100 | mV |
| | | $I_O$ = 5mA to 1A | Ð | 9.0 | 100 | |
| | | $I_O$ = 250mA to 750mA | Ð | 5.0 | 50.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | Ð | 4.3 | 6.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 19V to 25V, $I_O$ = 500mA | Ð | Ð | 0.8 | |
| | | $V_I$ = 8.5V to 21V, $T_J$ = +25¡C | Ð | Ð | 0.8 | |
| $V_O$/ T | Output Voltage Drift(22) | $I_O$ = 5mA | Ð | -0.8 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | Ð | 10.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection(22) | f = 120Hz, $I_O$ = 500mA, $V_I$ = 9V to 19V | Ð | 65.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance(22) | f = 1kHz | Ð | 17.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current(22) | $T_J$ = +25¡C | Ð | 2.2 | Ð | A |

Notes:

21. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
22. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7808A) (Continued)

Refer to the test circuits. 0¡C < $T_J$ < 125¡C, $I_O$ = 1A, $V_I$ = 14V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 7.84 | 8.0 | 8.16 | V |
| | | $I_O$ = 5mA to 1A, $P_O$ 15W, $V_I$ = 10.6V to 23V | | 7.7 | 8.0 | 8.3 | |
| Regline | Line Regulation(23) | $V_I$ = 10.6V to 25V, $I_O$ = 500mA | | Ð | 6.0 | 80.0 | mV |
| | | $V_I$ = 11V to 17V | | Ð | 3.0 | 80.0 | |
| | | $T_J$ = +25¡C | $V_I$ = 10.4V to 23V | Ð | 6.0 | 80.0 | |
| | | | $V_I$ = 11V to 17V | Ð | 2.0 | 40.0 | |
| Regload | Load Regulation(23) | $T_J$ = +25¡C, $I_O$ = 5mA to 1.5A | | Ð | 12.0 | 100 | mV |
| | | $I_O$ = 5mA to 1A | | Ð | 12.0 | 100 | |
| | | $I_O$ = 250mA to 750mA | | Ð | 5.0 | 50.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.0 | 6.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 11V to 25V, $I_O$ = 500mA | | Ð | Ð | 0.8 | |
| | | $V_I$ = 10.6V to 23V, $T_J$ = +25¡C | | Ð | Ð | 0.8 | |
| $V_O$/ T | Output Voltage Drift(24) | $I_O$ = 5mA | | Ð | -0.8 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 10.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection(24) | f = 120Hz, $I_O$ = 500mA, $V_I$ = 11.5V to 21.5V | | Ð | 62.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance(24) | f = 1kHz | | Ð | 18.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current(24) | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

23. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
24. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7809A) (Continued)

Refer to the test circuits. 0¡C < $T_J$ < 125¡C, $I_O$ = 1A, $V_I$ = 15V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | 8.82 | 9.0 | 9.16 | V |
| | | $I_O$ = 5mA to 1A, $P_O$ 15W, $V_I$ = 11.2V to 24V | 8.65 | 9.0 | 9.35 | |
| Regline | Line Regulation[25] | $V_I$ = 11.7V to 25V, $I_O$ = 500mA | Ð | 6.0 | 90.0 | mV |
| | | $V_I$ = 12.5V to 19V | Ð | 4.0 | 45.0 | |
| | | $T_J$ = +25¡C, $V_I$ = 11.5V to 24V | Ð | 6.0 | 90.0 | |
| | | $T_J$ = +25¡C, $V_I$ = 12.5V to 19V | Ð | 2.0 | 45.0 | |
| Regload | Load Regulation[25] | $T_J$ = +25¡C, $I_O$ = 5mA to 1.5A | Ð | 12.0 | 100 | mV |
| | | $I_O$ = 5mA to 1A | Ð | 12.0 | 100 | |
| | | $I_O$ = 250mA to 750mA | Ð | 5.0 | 50.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | Ð | 5.0 | 6.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 12V to 25V, $I_O$ = 500mA | Ð | Ð | 0.8 | |
| | | $V_I$ = 11.7V to 25V, $T_J$ = +25¡C | Ð | Ð | 0.8 | |
| $V_O$/ T | Output Voltage Drift[26] | $I_O$ = 5mA | Ð | -1.0 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | Ð | 10.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection[26] | f = 120Hz, $I_O$ = 500mA, $V_I$ = 12V to 22V | Ð | 62.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance[26] | f = 1kHz | Ð | 17.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current[26] | $T_J$ = +25¡C | Ð | 2.2 | Ð | A |

Notes:

25. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
26. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7810A) (Continued)

Refer to the test circuits. 0¡C < $T_J$ < 125¡C, $I_O$ = 1A, $V_I$ = 16V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | | 9.8 | 10.0 | 10.2 | V |
| | | $I_O$ = 5mA to 1A, $P_O$ 15W, $V_I$ = 12.8V to 25V | | 9.6 | 10.0 | 10.4 | |
| Regline | Line Regulation(27) | $V_I$ = 12.8V to 26V, $I_O$ = 500mA | | Ð | 8.0 | 100 | mV |
| | | $V_I$ = 13V to 20V | | Ð | 4.0 | 50.0 | |
| | | $T_J$ = +25¡C | $V_I$ = 12.5V to 25V | Ð | 8.0 | 100 | |
| | | | $V_I$ = 13V to 20V | Ð | 3.0 | 50.0 | |
| Regload | Load Regulation(27) | $T_J$ = +25¡C, $I_O$ = 5mA to 1.5A | | Ð | 12.0 | 100 | mV |
| | | $I_O$ = 5mA to 1A | | Ð | 12.0 | 100 | |
| | | $I_O$ = 250mA to 750mA | | Ð | 5.0 | 50.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | | Ð | 5.0 | 6.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 12.8V to 25V, $I_O$ = 500mA | | Ð | Ð | 0.8 | |
| | | $V_I$ = 13V to 26V, $T_J$ = +25¡C | | Ð | Ð | 0.5 | |
| $V_O$/ T | Output Voltage Drift(28) | $I_O$ = 5mA | | Ð | -1.0 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | | Ð | 10.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection(28) | f = 120Hz, $I_O$ = 500mA, $V_I$ = 14V to 24V | | Ð | 62.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance(28) | f = 1kHz | | Ð | 17.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current(28) | $T_J$ = +25¡C | | Ð | 2.2 | Ð | A |

Notes:

27. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
28. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7812A) (Continued)

Refer to the test circuits. 0¡C < $T_J$ < 125¡C, $I_O$ = 1A, $V_I$ = 19V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | 11.75 | 12.0 | 12.25 | V |
| | | $I_O$ = 5mA to 1A, $P_O$ 15W, $V_I$ = 14.8V to 27V | 11.5 | 12.0 | 12.5 | |
| Regline | Line Regulation(29) | $V_I$ = 14.8V to 30V, $I_O$ = 500mA | Ð | 10.0 | 120 | mV |
| | | $V_I$ = 16V to 22V | Ð | 4.0 | 120 | |
| | | $T_J$ = +25¡C, $V_I$ = 14.5V to 27V | Ð | 10.0 | 120 | |
| | | $T_J$ = +25¡C, $V_I$ = 16V to 22V | Ð | 3.0 | 60.0 | |
| Regload | Load Regulation(29) | $T_J$ = +25¡C, $I_O$ = 5mA to 1.5A | Ð | 12.0 | 100 | mV |
| | | $I_O$ = 5mA to 1A | Ð | 12.0 | 100 | |
| | | $I_O$ = 250mA to 750mA | Ð | 5.0 | 50.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | Ð | 5.1 | 6.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 14V to 27V, $I_O$ = 500mA | Ð | Ð | 0.8 | |
| | | $V_I$ = 15V to 30V, $T_J$ = +25¡C | Ð | Ð | 0.8 | |
| $V_O$/ T | Output Voltage Drift(30) | $I_O$ = 5mA | Ð | -1.0 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | Ð | 10.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection(30) | f = 120Hz, $I_O$ = 500mA, $V_I$ = 14V to 24V | Ð | 60.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance(30) | f = 1kHz | Ð | 18.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current(30) | $T_J$ = +25¡C | Ð | 2.2 | Ð | A |

Note:

29. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
30. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7815A) (Continued)

Refer to the test circuits. 0¡C < $T_J$ < 125¡C, $I_O$ = 1A, $V_I$ = 23V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | 14.75 | 15.0 | 15.3 | V |
| | | $I_O$ = 5mA to 1A, $P_O$ 15W, $V_I$ = 17.7V to 30V | 14.4 | 15.0 | 15.6 | |
| Regline | Line Regulation(31) | $V_I$ = 17.4V to 30V, $I_O$ = 500mA | Ð | 10.0 | 150 | mV |
| | | $V_I$ = 20V to 26V | Ð | 5.0 | 150 | |
| | | $T_J$ = +25¡C, $V_I$ = 17.5V to 30V | Ð | 11.0 | 150 | |
| | | $T_J$ = +25¡C, $V_I$ = 20V to 26V | Ð | 3.0 | 75.0 | |
| Regload | Load Regulation(31) | $T_J$ = +25¡C, $I_O$ = 5mA to 1.5A | Ð | 12.0 | 100 | mV |
| | | $I_O$ = 5mA to 1A | Ð | 12.0 | 100 | |
| | | $I_O$ = 250mA to 750mA | Ð | 5.0 | 50.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | Ð | 5.2 | 6.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 17.5V to 30V, $I_O$ = 500mA | Ð | Ð | 0.8 | |
| | | $V_I$ = 17.5V to 30V, $T_J$ = +25¡C | Ð | Ð | 0.8 | |
| $V_O$/ T | Output Voltage Drift(32) | $I_O$ = 5mA | Ð | -1.0 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | Ð | 10.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection(32) | f = 120Hz, $I_O$ = 500mA, $V_I$ = 18.5V to 28.5V | Ð | 58.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance(32) | f = 1kHz | Ð | 19.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current(32) | $T_J$ = +25¡C | Ð | 2.2 | Ð | A |

Notes:

31. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
32. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7818A) (Continued)

Refer to the test circuits. 0¡C < $T_J$ < 125¡C, $I_O$ = 1A, $V_I$ = 27V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | 17.64 | 18.0 | 18.36 | V |
| | | $I_O$ = 5mA to 1A, $P_O$ 15W, $V_I$ = 21V to 33V | 17.3 | 18.0 | 18.7 | |
| Regline | Line Regulation[33] | $V_I$ = 21V to 33V, $I_O$ = 500mA | Ð | 15.0 | 180 | mV |
| | | $V_I$ = 21V to 33V | Ð | 5.0 | 180 | |
| | | $T_J$ = +25¡C, $V_I$ = 20.6V to 33V | Ð | 15.0 | 180 | |
| | | $T_J$ = +25¡C, $V_I$ = 24V to 30V | Ð | 5.0 | 90.0 | |
| Regload | Load Regulation[33] | $T_J$ = +25¡C, $I_O$ = 5mA to 1.5A | Ð | 15.0 | 100 | mV |
| | | $I_O$ = 5mA to 1A | Ð | 15.0 | 100 | |
| | | $I_O$ = 250mA to 750mA | Ð | 7.0 | 50.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | Ð | 5.2 | 6.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 12V to 33V, $I_O$ = 500mA | Ð | Ð | 0.8 | |
| | | $V_I$ = 12V to 33V, $T_J$ = +25¡C | Ð | Ð | 0.8 | |
| $V_O$/ T | Output Voltage Drift[34] | $I_O$ = 5mA | Ð | -1.0 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | Ð | 10.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection[34] | f = 120Hz, $I_O$ = 500mA, $V_I$ = 22V to 32V | Ð | 57.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance[34] | f = 1kHz | Ð | 19.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current[34] | $T_J$ = +25¡C | Ð | 2.2 | Ð | A |

Notes:

33. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
34. These parameters, although guaranteed, are not 100% tested in production.

## Electrical Characteristics (LM7824A) (Continued)

Refer to the test circuits. 0¡C < $T_J$ < 125¡C, $I_O$ = 1A, $V_I$ = 33V, $C_I$ = 0.33µF, $C_O$ = 0.1µF, unless otherwise speciÞed.

| Symbol | Parameter | Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $V_O$ | Output Voltage | $T_J$ = +25¡C | 23.5 | 24.0 | 24.5 | V |
| | | $I_O$ = 5mA to 1A, $P_O$ 15W, $V_I$ = 27.3V to 38V | 23.0 | 24.0 | 25.0 | |
| Regline | Line Regulation[35] | $V_I$ = 27V to 38V, $I_O$ = 500mA | Ð | 18.0 | 240 | mV |
| | | $V_I$ = 21V to 33V | Ð | 6.0 | 240 | |
| | | $T_J$ = +25¡C $V_I$ = 26.7V to 38V | Ð | 18.0 | 240 | |
| | | $V_I$ = 30V to 36V | Ð | 6.0 | 120 | |
| Regload | Load Regulation[35] | $T_J$ = +25¡C, $I_O$ = 5mA to 1.5A | Ð | 15.0 | 100 | mV |
| | | $I_O$ = 5mA to 1A | Ð | 15.0 | 100 | |
| | | $I_O$ = 250mA to 750mA | Ð | 7.0 | 50.0 | |
| $I_Q$ | Quiescent Current | $T_J$ = +25¡C | Ð | 5.2 | 6.0 | mA |
| $I_Q$ | Quiescent Current Change | $I_O$ = 5mA to 1A | Ð | Ð | 0.5 | mA |
| | | $V_I$ = 27.3V to 38V, $I_O$ = 500mA | Ð | Ð | 0.8 | |
| | | $V_I$ = 27.3V to 38V, $T_J$ = +25¡C | Ð | Ð | 0.8 | |
| $V_O$/ T | Output Voltage Drift[36] | $I_O$ = 5mA | Ð | -1.5 | Ð | mV/¡C |
| $V_N$ | Output Noise Voltage | f = 10Hz to 100kHz, $T_A$ = +25¡C | Ð | 10.0 | Ð | µV/$V_O$ |
| RR | Ripple Rejection[36] | f = 120Hz, $I_O$ = 500mA, $V_I$ = 28V to 38V | Ð | 54.0 | Ð | dB |
| $V_{DROP}$ | Dropout Voltage | $I_O$ = 1A, $T_J$ = +25¡C | Ð | 2.0 | Ð | V |
| $r_O$ | Output Resistance[36] | f = 1kHz | Ð | 20.0 | Ð | m |
| $I_{SC}$ | Short Circuit Current | $V_I$ = 35V, $T_A$ = +25¡C | Ð | 250 | Ð | mA |
| $I_{PK}$ | Peak Current[36] | $T_J$ = +25¡C | Ð | 2.2 | Ð | A |

Notes:

35. Load and line regulation are speciÞed at constant junction temperature. Changes in $V_O$ due to heating effects must be taken into account separately. Pulse testing with low duty is used.
36. These parameters, although guaranteed, are not 100% tested in production.

## Typical Performance Characteristics

Figure 3. Quiescent Current

Figure 4. Peak Output Current

Figure 5. Output Voltage

Figure 6. Quiescent Current

## Typical Applications

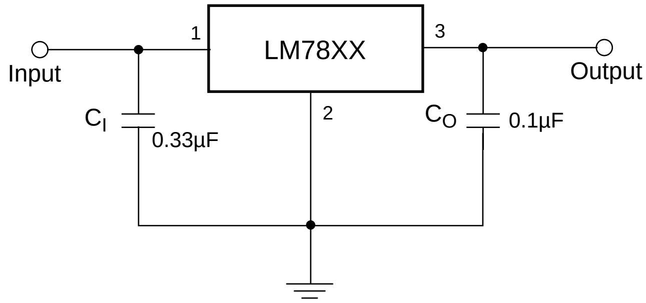



Figure 7. DC Parameters

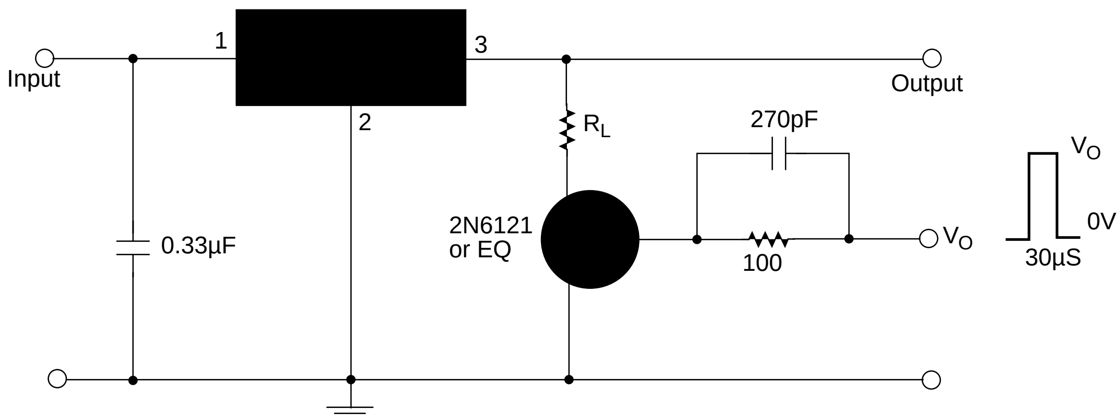



Figure 8. Load Regulation

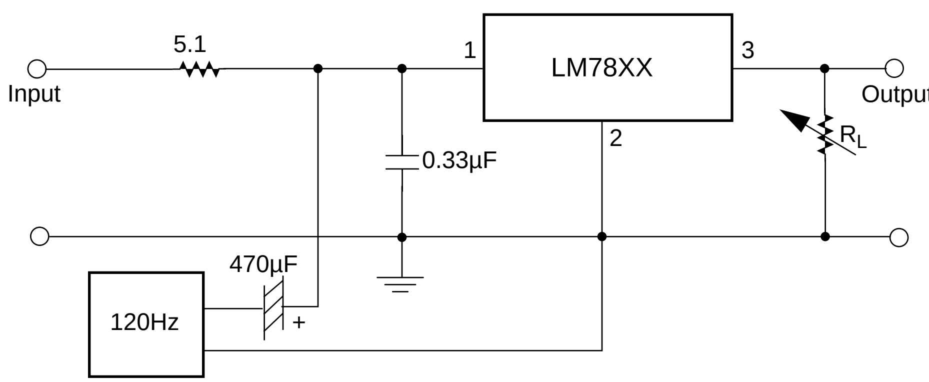



Figure 9. Ripple Rejection

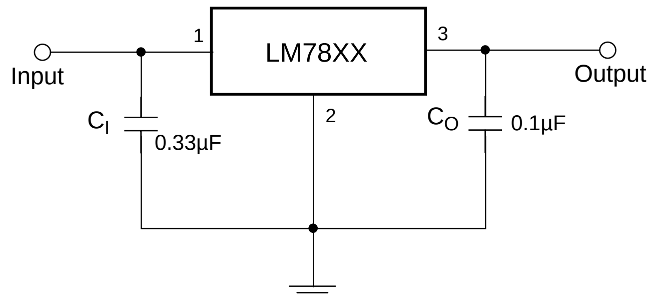
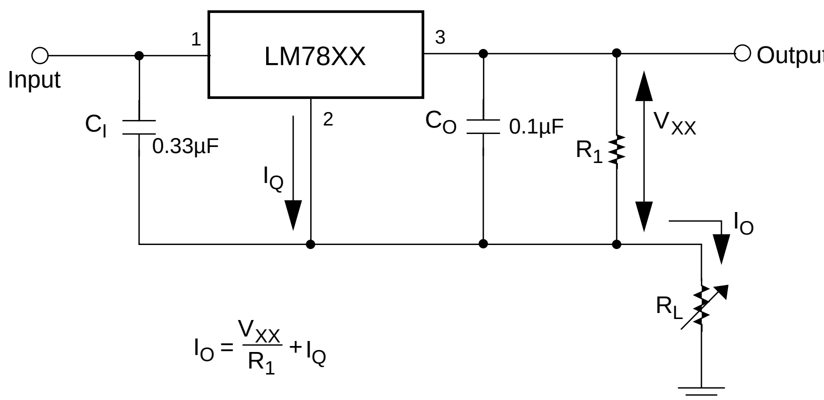



Figure 10. Fixed Output Regulator

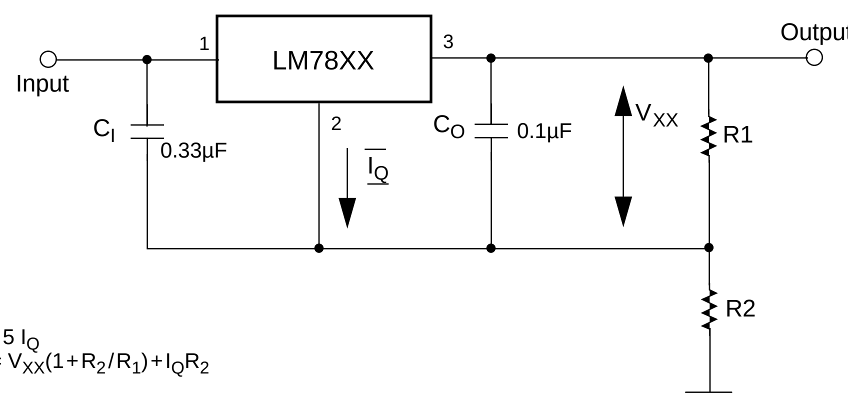



$$I_O = \frac{V_{XX}}{R_1} + I_Q$$

Notes:

1. To specify an output voltage, substitute voltage value for ÒXX.Ó A common ground is required between the input and the output voltage. The input voltage must remain typically 2.0V above the output voltage even during the low point on the input ripple voltage.
2. $C_I$ is required if regulator is located an appreciable distance from power supply Þlter.
3. $C_O$ improves stability and transient response.

Figure 11.




$I_{RI}$ 5 $I_Q$
$V_O = V_{XX}(1+R_2/R_1)+I_QR_2$

Figure 12. Circuit for Increasing Output Voltage

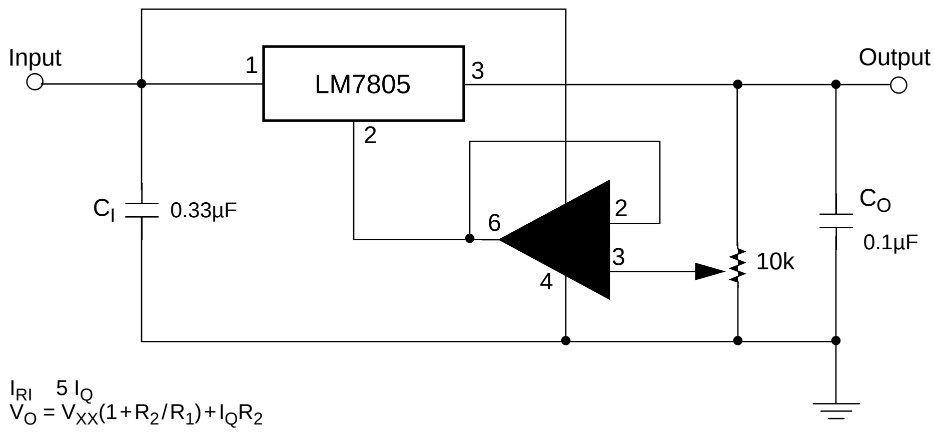



Figure 13. Adjustable Output Regulator (7V to 30V)

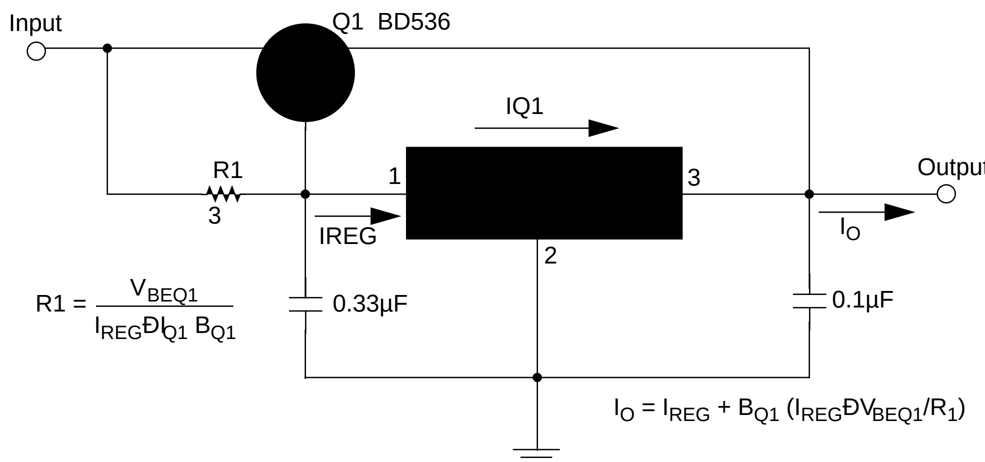



Figure 14. High Current Voltage Regulator

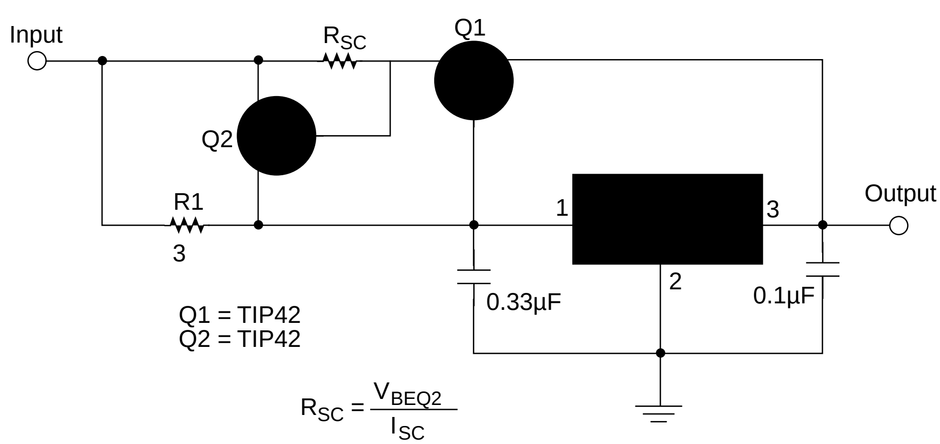



Figure 15. High Output Current with Short Circuit Protection

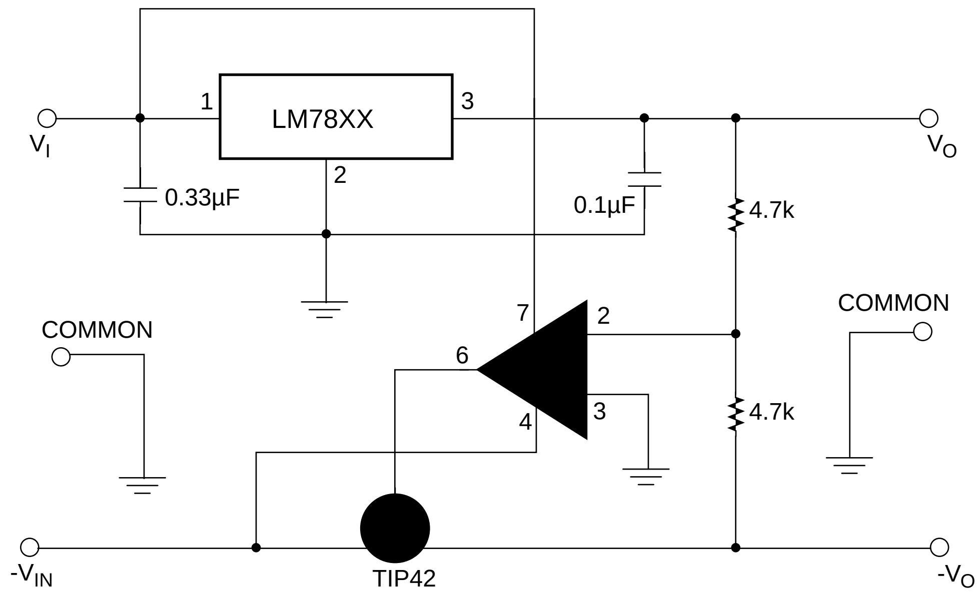



Figure 16. Tracking Voltage Regulator

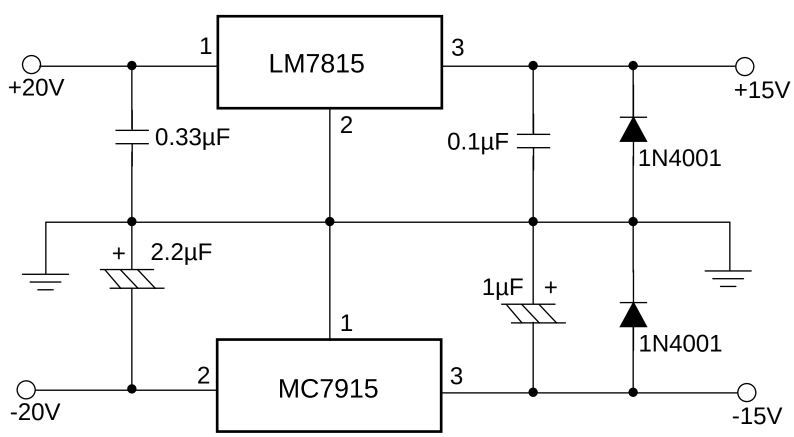



Figure 17. Split Power Supply (±15V Ð 1A)

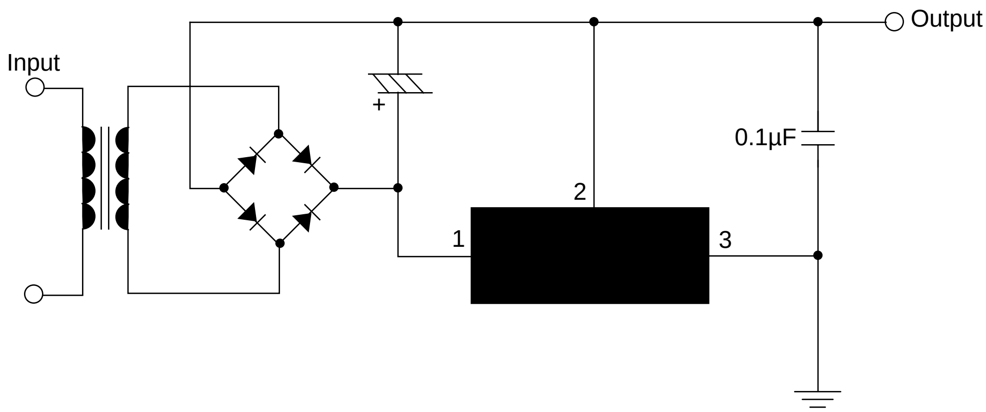



Figure 18. Negative Output Voltage Circuit

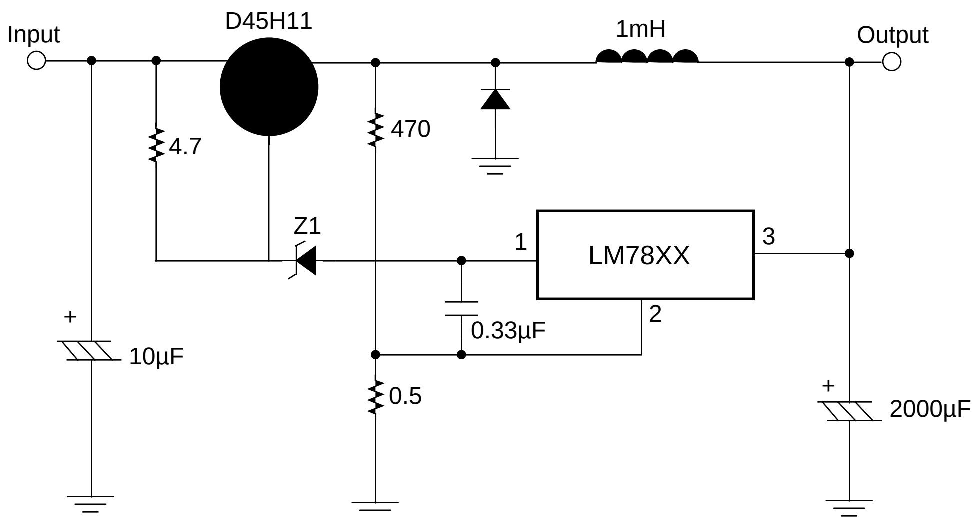
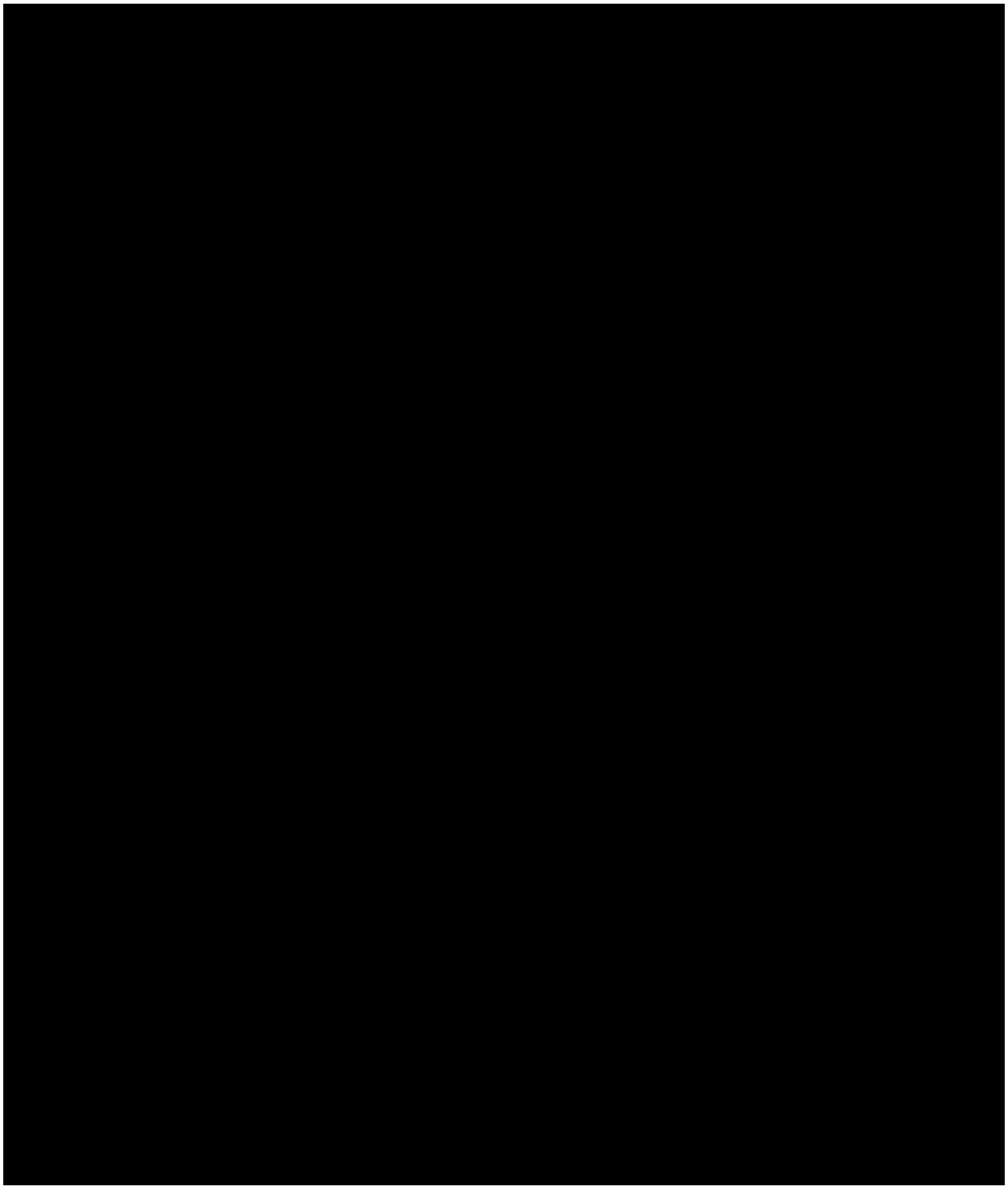



Figure 19. Switching Regulator

## Mechanical Dimensions

Dimensions in millimeters

### TO-220



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